

Goals for today

- Admin. items:
 - Upcoming deadlines:
 - **Tomorrow** at midnight: Pset 1
 - Oct. 19: Project proposal
 - Oct. 26: Pset 2
 - Our first paper presentation is on Wednesday (Eva, Antara, & Aswin)
 - I posted links to 2 'low-overhead' software libraries on the Canvas *Additional resources* page
- Complete forward kinematics
- Start differential kinematics

Review of key concepts from last week

We define the screw axis $\mathcal{S} = \begin{bmatrix} \omega \\ v \end{bmatrix} \in \mathbb{R}^6$ using a normalized version of any twist $\mathcal{V} = (\omega, v)$ corresponding to motion along the screw:

1. If $\omega \neq 0$, $\mathcal{S} = \mathcal{V}/\|\omega\| = (\omega/\|\omega\|, v/\|\omega\|)$
2. If $\omega = 0$, $\mathcal{S} = \mathcal{V}/\|v\| = (0, v/\|v\|)$

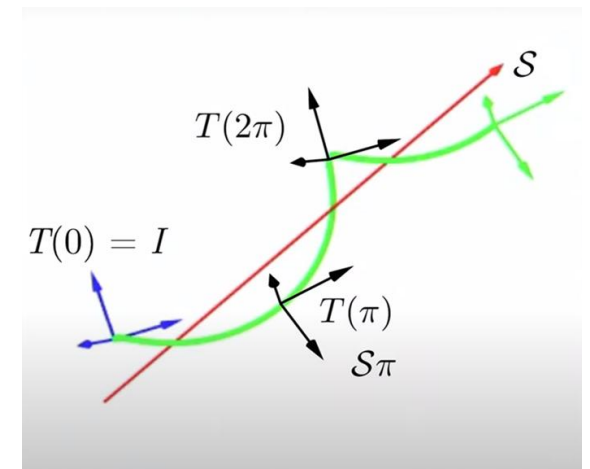
The 4 x 4 matrix representation $S = (\omega, v)$ is $[S] = \begin{bmatrix} [\omega] & v \\ 0 & 0 \end{bmatrix}$

We can transform screw axes using the adjoint map $\mathcal{S}_s = [\text{Ad}_{T_{sb}}]\mathcal{S}_b$ where $[\text{Ad}_T] = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \in \mathbb{R}^{6 \times 6}$

We parameterize a homogeneous transformation with exponential coordinates:

$$e^{[S]\theta} = \begin{bmatrix} e^{[\omega]\theta} & (I\theta + (1 - \cos(\theta))[\omega] + (\theta - \sin(\theta))[\omega]^2)v \\ 0 & 1 \end{bmatrix} \text{ if } \|\omega\| = 1$$

$$e^{[S]\theta} = \begin{bmatrix} I & v\theta \\ 0 & 1 \end{bmatrix} \text{ if } \omega = 0 \text{ and } \|v\| = 1$$



Product of Exponentials: Space form

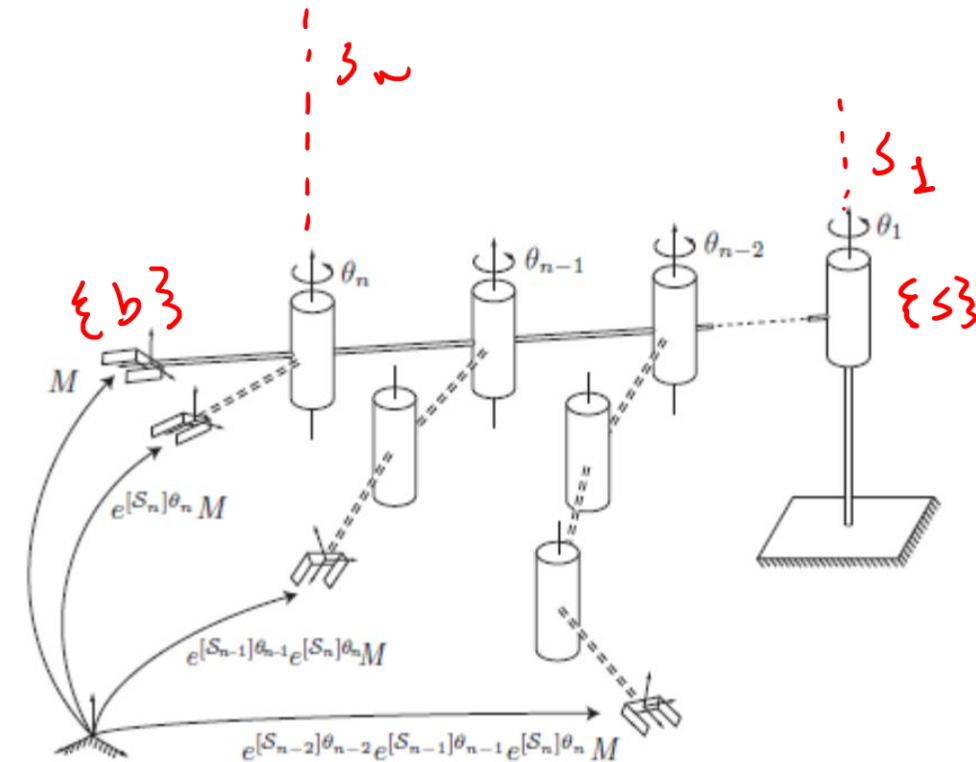
Step 1

- Choose a fixed base frame $\{s\}$ and an end-effector frame attached to the last link $\{b\}$
- Set all joint angles to 0
- Find $M \in SE(3)$, the configuration of $\{b\}$ relative to $\{s\}$ when the value of all joints are set to 0

Step 2

- Suppose joint n is displaced to some joint value θ_n . This will lead to a transformation of $\{b\}$ where $T = e^{[S_n]\theta_n}M$ and S_n is the screw axis of joint n in the $\{s\}$ frame
- Complete this for $n, n-1, \dots, 1$ until you have

$$T_{sb}(\theta_1, \dots, \theta_n) = e^{[S_1]\theta_1} \dots e^{[S_n]\theta_n} M$$



Product of Exponentials: Body form

It can also be useful to express the results in the equivalent **body form**. Starting from:

$$T(\theta) = e^{[S_1]\theta_1} \dots e^{[S_n]\theta_n} M$$

Using the matrix identity $e^{M^{-1}PM} = M^{-1}e^P M$ therefore $Me^{M^{-1}PM} = e^P M$:

$$T(\theta) = e^{[S_1]\theta_1} \dots Me^{M^{-1}[S_n]M\theta_n}$$

We keep using the identity for each step until we find:

$$T(\theta) = Me^{[B_1]\theta_1} \dots e^{[B_n]\theta_n}, \text{ where } [B_i] = M^{-1}[S_i]M \quad (\text{i.e. } B_i = [\text{Ad}_{M^{-1}}]S_i)$$

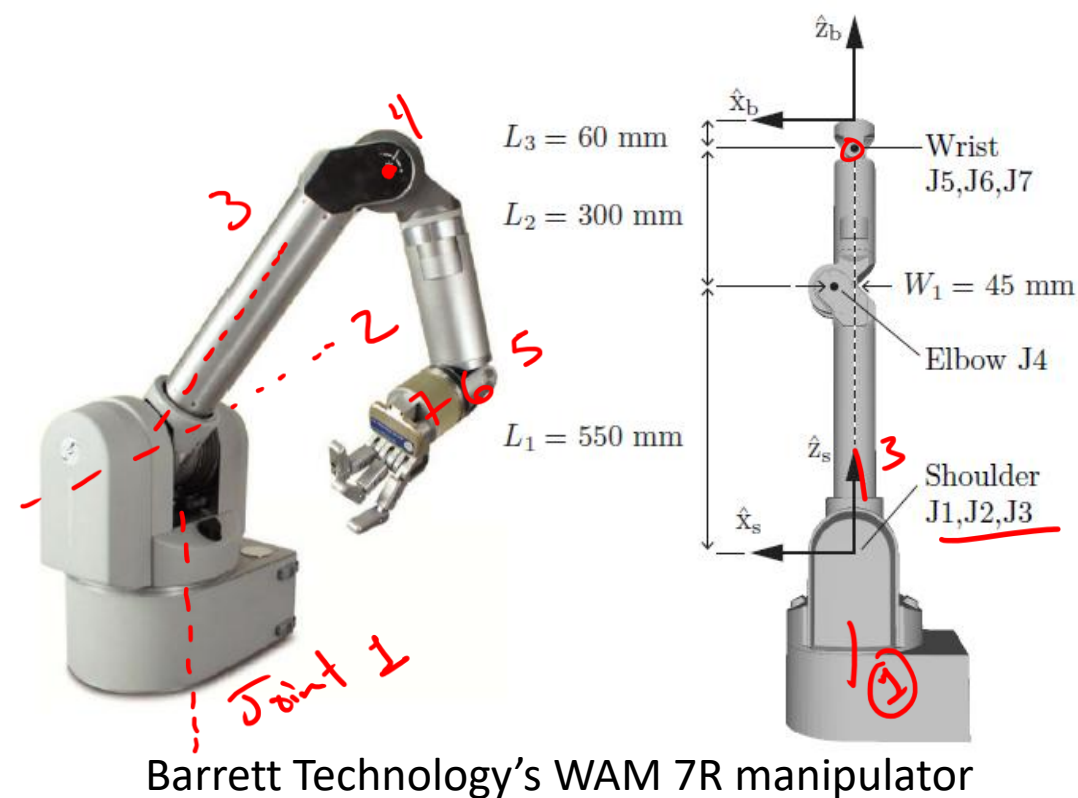
In **body form**, $B_i = (\omega_i, v_i)$ and is the screw axis corresponding to the joint axis i expressed in the end-effector, or body, frame $\{b\}$, with the robot in the zero position.

Space form vs. body form

<u>Space form</u>	<u>Body form</u>
• Screw axes expressed in $\{s\}$ frame • M is first transformed by the most distal joint (n), progressively moving inward to proximal joints • <i>Joint i doesn't affect joint $i - 1$'s screw axis in the $\{s\}$ frame</i>	• Screw axes expressed in $\{b\}$ frame • M is first transformed by the most proximal joint (1^{st}), progressively moving outward to distal joints • <i>Joint $i - 1$ doesn't affect joint i's screw axis in the $\{b\}$ frame</i>

We only need to determine the screw axes at the robot's zero position: any S_i is unaffected by more distal transformations, and any B_i is unaffected by more proximal transformations.

Example



Barrett WAM manipulator

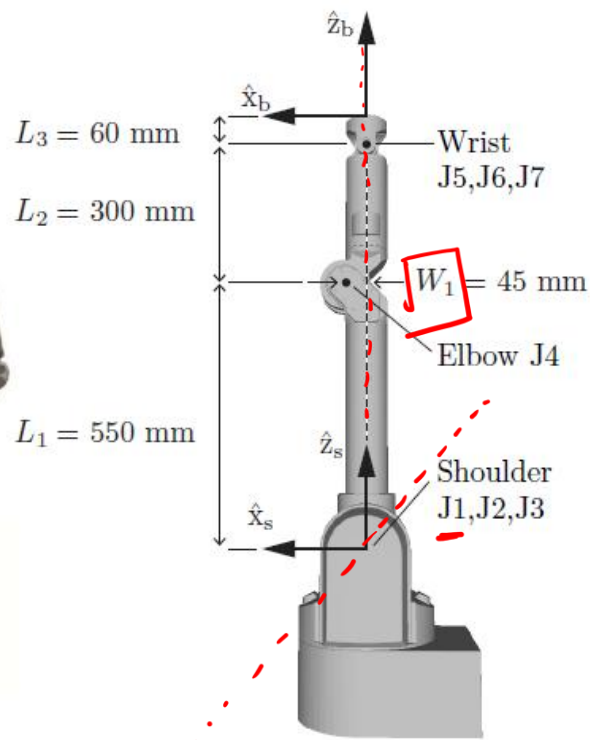
- redundant; has 7 DOF
- some joints driven by motors in the base using cable transmissions (contrast the robot w/ the UR5 where motors are at the joints)

At the zero configuration:

- axes 1, 3, 5, 7 are along $\hat{z}_s$
- axes 2, 4, 6 are aligned w/ $\hat{y}_s$ out of page
- $J1, J2, J3$ intersect @ origin of $\{s\}$
- $J5, J6, J7$ intersect @ a point 60 mm from $\{b\}$

At the home position:

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & L_1 + L_2 + L_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Find the screw axes in the $\{b\}$ frame

$$x \in \mathbb{R}^3 \Rightarrow \text{skew-symmetric } [x] = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}$$

$$\omega_1 = [0, 0, 1]$$

$$v_1 = -\omega_1 \times q_1 = \begin{bmatrix} 0 & +1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = [0, 0, 0]$$

$$\omega_2 = [0, 1, 0]$$

$$v_2 = -\omega_2 \times q_2 = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -(L_1 + L_2 + L_3) \end{bmatrix} = \begin{bmatrix} L_1 + L_2 + L_3 \\ 0 \\ 0 \end{bmatrix}$$

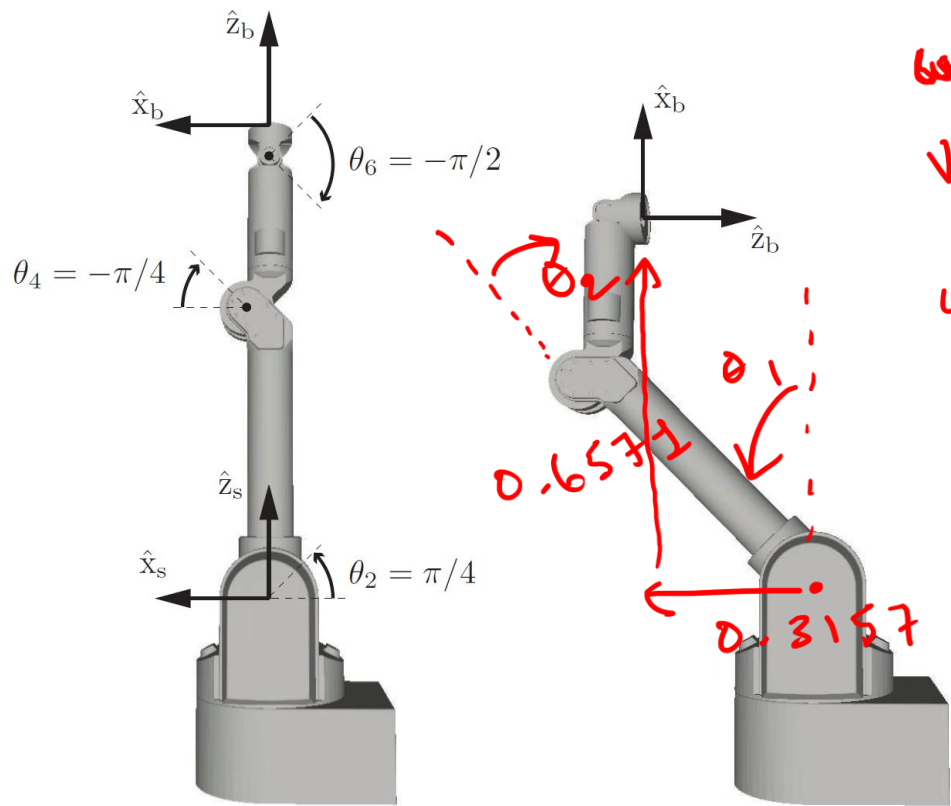
$$\omega_3 = [0, 0, 2]$$

$$v_3 = [0, 0, 0]$$

$$\omega_4 = [0, 1, 0]$$

$$v_4 = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \omega_1 \\ 0 \\ -(L_2 + L_3) \end{bmatrix} = [L_2 + L_3, 0, \omega_1]$$

Need to include elbow offset



$$\omega_5 = [0, 0, 1]$$

$$v_5 = [0, 0, 0]$$

$$\omega_6 = [0, 1, 0]$$

$$v_6 = [L_3, 0, 0]$$

$$\omega_7 = [0, 0, 1]$$

$$v_7 = [0, 0, 0]$$

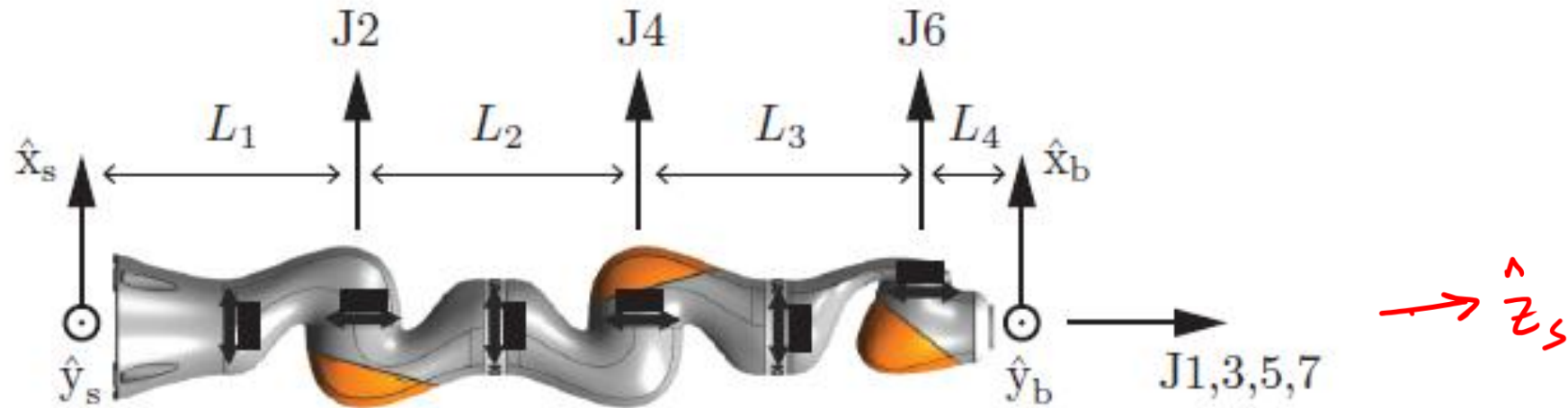
The figure shows the WAM arm w/
 $\theta_2 = 45^\circ$, $\theta_4 = -45^\circ$, $\theta_6 = -90^\circ$ and all other
 joints equal to zero.

$$T_{sb}(\theta) = M e^{[B_2]\pi/4} e^{-[B_4]\pi/4} e^{-[B_6]\pi/2}$$

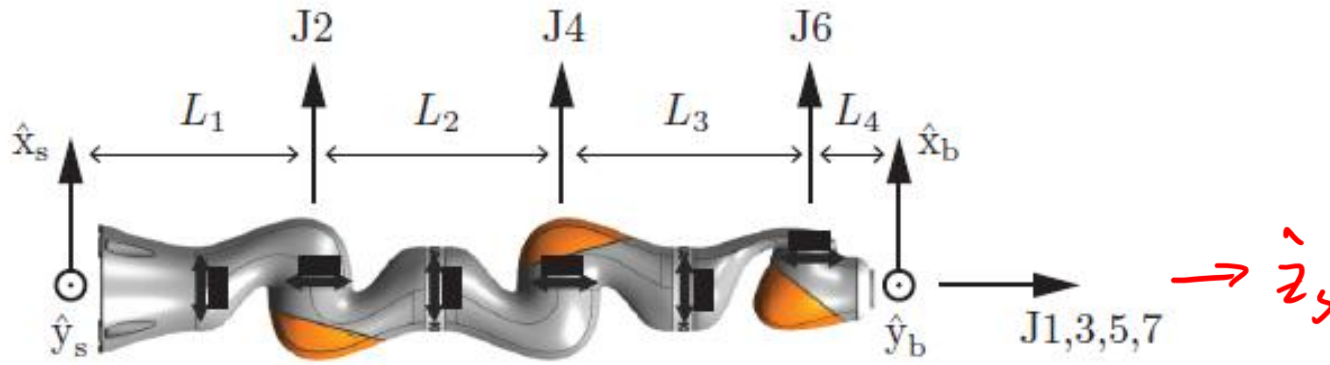
$$= \begin{bmatrix} 0 & 0 & -1 & 0.3157 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0.6571 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Solved w/
 eqn. 3.88

Example

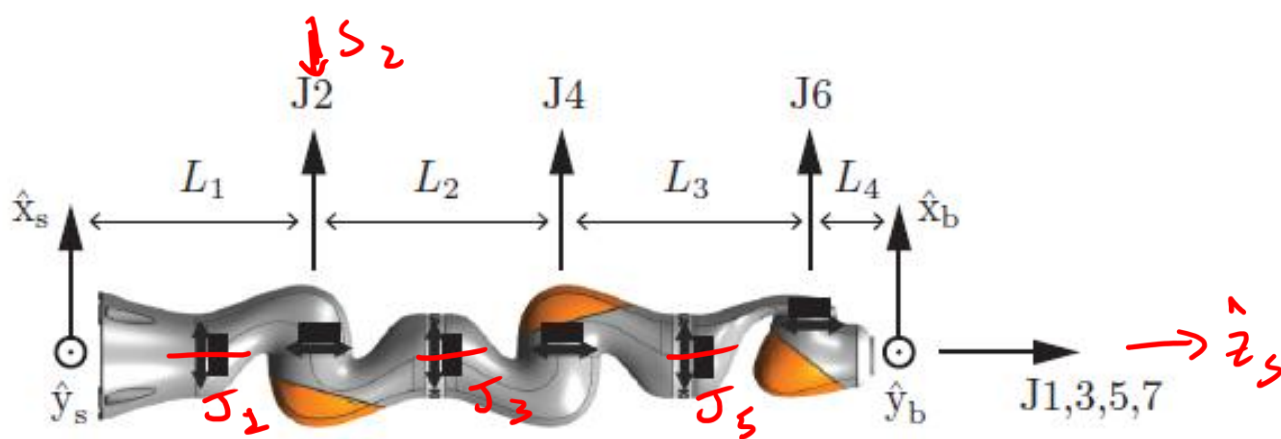


- The figure above shows the KUKA LBR iiwa (LBR = “Leichtbauroboter,” German for lightweight robot; iiwa = “intelligent industrial work assistant”) 7R robot arm.
- The figure defines an $\{s\}$ frame at the base with the $\hat{y}_s$ -axis pointing out of the page and a $\{b\}$ frame aligned with $\{s\}$ at the end-effector.
- The robot is at its home configuration. The screw axes for the seven joints are illustrated (positive rotation about these axes is by the righthand rule).
- The axes for joints 2, 4, and 6 are aligned, and the axes for joints 1, 3, 5, and 7 are identical at the home configuration.

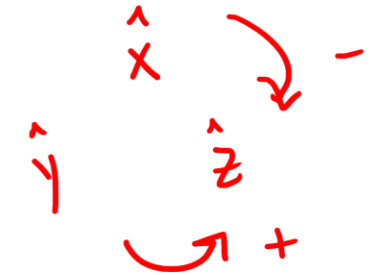


Part a) Write M

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & L_1 + L_2 + L_3 + L_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Part b) Write the screw axes in {s}



$$\omega_1 = [0 \ 0 \ 1]; \quad v_1 = [0 \ 0 \ 0]$$

$$\omega_2 = [1 \ 0 \ 0]$$

$$v_2 = - \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ L_1 \end{bmatrix} = [0 \ L_1 \ 0]$$

$$\omega_3 = [0 \ 0 \ 1]; \quad v_3 = [0 \ 0 \ 0]$$

$$\omega_4 = [1 \ 0 \ 0]; \quad v_4 = [0 \ (L_1 + L_2) \ 0]$$

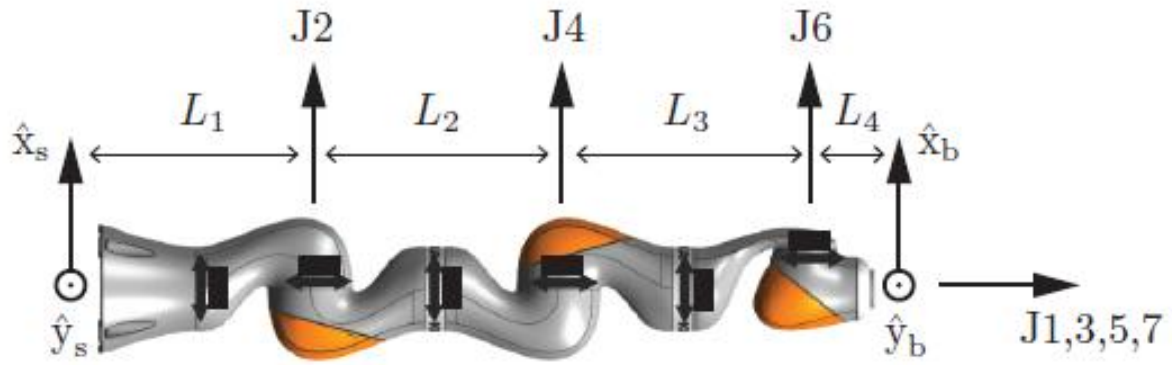
$$\omega_5 = [0 \ 0 \ 1]; \quad v_5 = [0 \ 0 \ 0]$$

$$\omega_6 = [1 \ 0 \ 0]$$

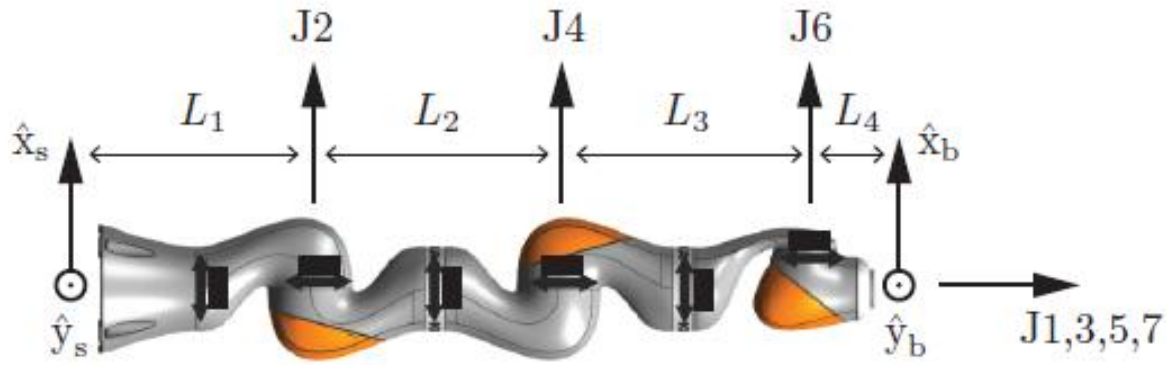
$$v_6 = [0 \ (L_1 + L_2 + L_3) \ 0]$$

$$\omega_7 = [0 \ 0 \ 1]$$

$$v_7 = [0 \ 0 \ 0]$$



$$S = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & L_1 & 0 & L_1 + L_2 & 0 & L_1 + L_2 + L_3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$



Part c) Write the screw axes in {b}

$$\omega_1 = \omega_3 = \omega_5 = \omega_7 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

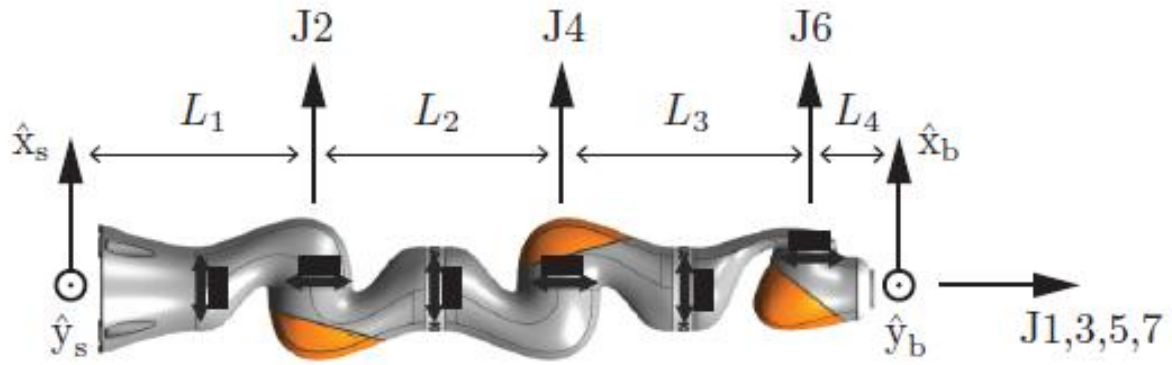
$$v_1 = v_3 = v_5 = v_7 = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\omega_2 = \omega_4 = \omega_6 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 0 & -(L_2 + L_3 + L_4) & 0 \end{bmatrix}$$

$$v_4 = \begin{bmatrix} 0 & -(L_3 + L_4) & 0 \end{bmatrix}$$

$$v_6 = \begin{bmatrix} 0 & -L_4 & 0 \end{bmatrix}$$



$$B = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -(L_2 + L_3 + L_4) & 0 & 0 & -(L_3 + L_4) & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

clear all

% Forward kinematics of the UR5 (see pg. 145 of Modern Robotics)

L1 = 0.425;

L2 = 0.392;

W1 = 0.109;

W2 = 0.082;

H1 = 0.089;

H2 = 0.095;

%

thetalist = [-pi/3; -pi/4; -pi/2; pi/8; pi/4; 0];

%

M = [[-1, 0, 0, L1+L2]; [0, 0, 1, W1+W2]; [0, 1, 0, H1-H2]; [0, 0, 0, 1]];

Slist = [[0; 0; 1; 0; 0; 0],...

[0; 1; 0; -H1; 0; 0],...

[0; 1; 0; -H1; 0; L1],...

[0; 1; 0; -H1; 0; L1+L2],...

[0; 0; -1; -W1; L1+L2; 0],...

[0; 1; 0; H2-H1; 0; L1+L2]

];

TS = FKinSpace(M, Slist, thetalist) % Modern Robotics function

TS =

0.7477	-0.4619	0.4771	0.1891
0.1192	0.8001	0.5879	0.0065
-0.6533	-0.3827	0.6533	0.7566
0	0	0	1.0000

% convert screw axes to body frame -> adjoint map(M^{-1})

Blist = Adjoint(inv(M))*Slist; % vectorized

TB = FKinBody(M, Blist, thetalist) % Modern Robotics function

TB =

0.7477	-0.4619	0.4771	0.1891
0.1192	0.8001	0.5879	0.0065
-0.6533	-0.3827	0.6533	0.7566
0	0	0	1.0000

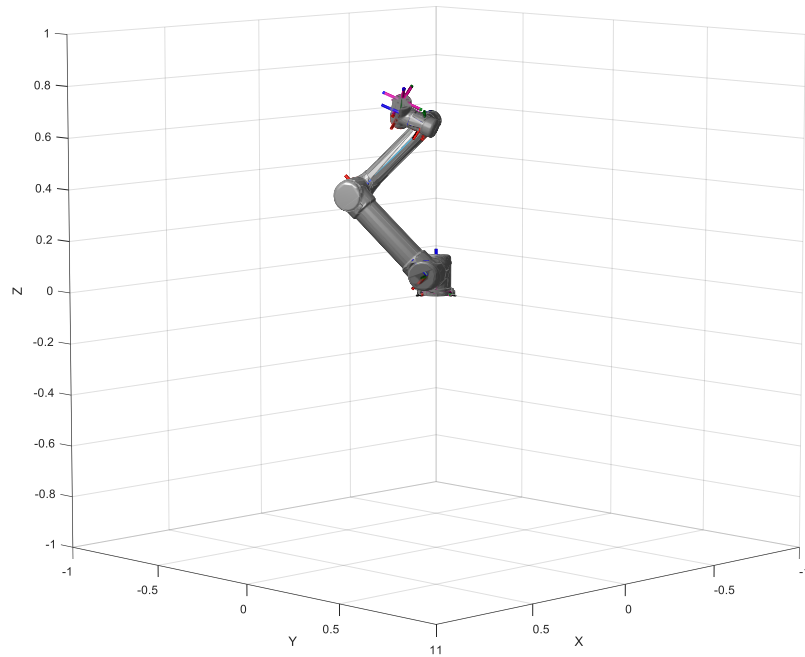
% the following visualizations use the Matlab Robotics System Toolbox

```
ur5 = loadrobot("universalUR5");
```

```
ur5.DataFormat = 'column';
```

```
show(ur5,thetalist)
```

```
thetalist = [-pi/3; -pi/4; -pi/2; pi/8; pi/4; 0];
```



```
thetalist = [pi/3; pi/4; -pi/2; pi/8; pi/4; 0];
```

